EEG report classification with MedGemma-27B — summary

Turning free-text clinical EEG reports into five structured diagnostic labels (overall abnormality; focal / generalized epileptiform; focal / generalized non-epileptiform) with a small, CPU-run, grammar-constrained **MedGemma-27B**. Benchmarked against the paper's **Mistral-7B** (Tian et al.) and a **human** second annotator (SG), on 1994 reports (two neurologists, "Zoe" + "Maria"), scored against reference annotator LD.

> **Key result:** our best prompt, **v5**, reaches **87.6%** whole-report accuracy — it **beats Mistral-7B on all five categories** and comes within ~2 points of the human annotator (89.8%).

Where each version lands

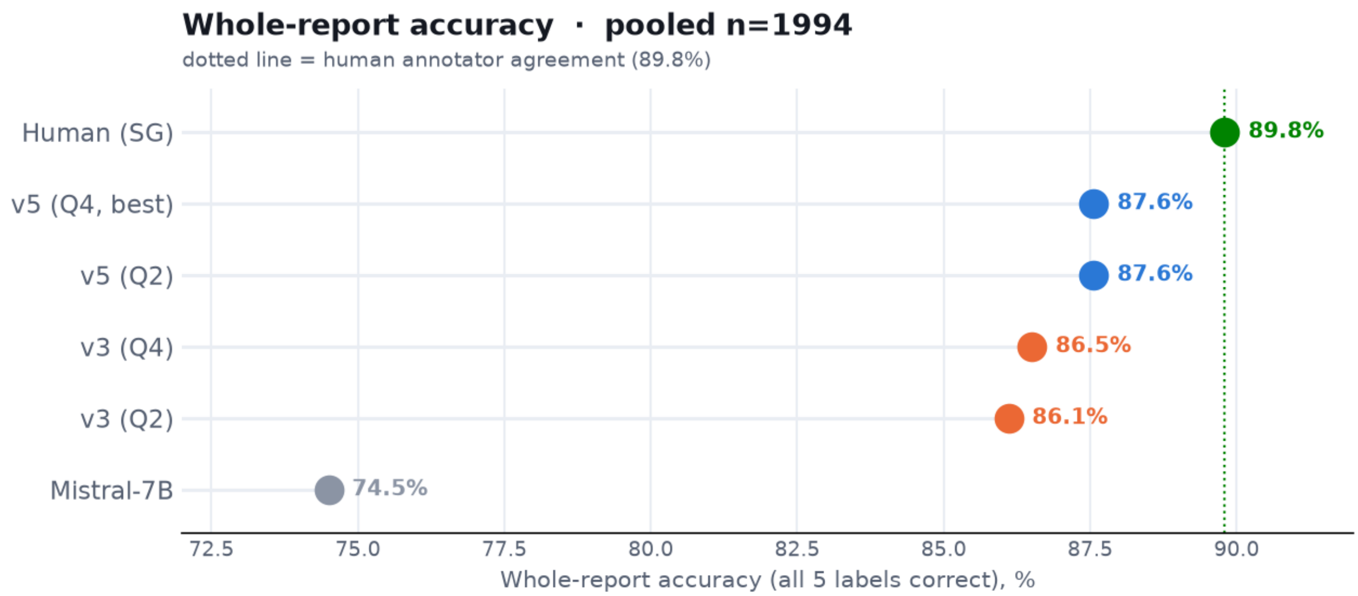

Figure 1: Whole-report accuracy

Two prompt generations, at both model sizes (Q2 ≈ 10 GB, Q4 ≈ 15 GB): **v3** adds an explicit rule for the rare *focal epileptiform* class; **v5** adds a *focal-vs-generalized* rule for slowing. Each step climbs toward the human line; **v5 (Q4)** is the best.

Per category

v5 (blue) beats Mistral (grey) in every category — by 10–15 points on the harder three (Gen Epi, Focal Non-epi, Gen Non-epi) — and lands **at the human level on the epileptiform categories** (Focal Epi, Gen Epi), trailing the human only on Abnormality and focal slowing.

Numbers (pooled n = 1994, Core F1 vs LD)

Model	Abnorm	Focal Epi	Gen Epi	Focal Non	Gen Non	Whole report
Mistral-7B	94.7	82.8	74.8	75.6	75.2	74.5
v3 (Q2)	95.2	88.5	90.4	88.2	85.3	86.1
v3 (Q4)	95.9	86.4	89.2	86.9	89.6	86.5
v5 (Q2)	95.4	87.6	89.9	88.8	86.7	87.6
v5 (Q4, best)	96.0	85.6	88.8	89.0	89.3	**87.6**
Human (SG)	98.0	85.7	87.5	90.8	90.0	89.8

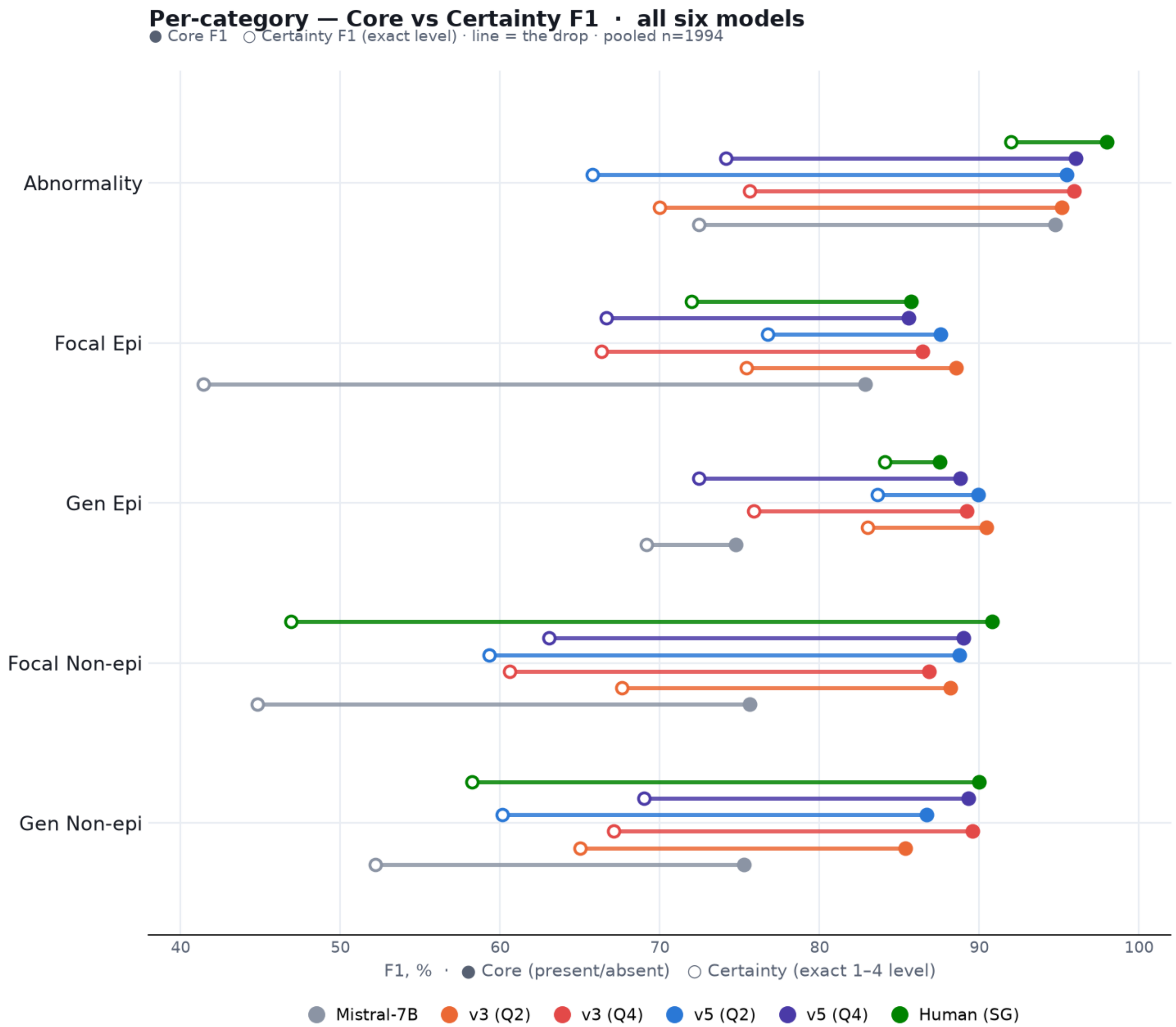

Figure 2: Per-category F1

Per-category values are F1 (0–100), which measures how well each finding is caught — the standard metric for these imbalanced classes. "Whole report" is the share of reports with all five labels correct.

Confidence — Certainty (exact-level) F1

The **Certainty** metric is stricter: the model must also get the exact 1–4 confidence level right. This is where our approach separates most from Mistral — on the per-category chart above it is the **open circle** (the drop from the filled Core dot).

Pooled (n = 1994):

Model	Abnorm	Focal Epi	Gen Epi	Focal Non	Gen Non
Mistral-7B	72.4	41.4	69.2	44.8	52.2
v3 (Q2)	70.0	75.4	83.0	67.6	65.0
v3 (Q4)	75.6	66.3	75.9	60.6	67.1
v5 (Q2)	65.8	76.8	83.6	59.3	60.1
v5 (Q4)	74.1	66.7	72.4	63.1	69.0
Human (SG)	92.0	72.0	84.1	46.9	58.3

Zoe (n = 1495):

Model	Abnorm	Focal Epi	Gen Epi	Focal Non	Gen Non
Mistral-7B	72.5	42.7	65.5	41.4	55.0
v3 (Q2)	67.7	69.9	82.2	66.7	67.1
v3 (Q4)	77.3	64.7	73.7	64.3	71.8
v5 (Q2)	61.9	72.6	83.0	53.7	60.8
v5 (Q4)	74.7	63.8	70.6	65.8	72.2
Human (SG)	91.5	69.8	84.2	42.5	58.9

Maria (n = 499):

Model	Abnorm	Focal Epi	Gen Epi	Focal Non	Gen Non
Mistral-7B	72.2	38.8	83.7	55.5	36.5
v3 (Q2)	77.9	86.7	85.7	70.2	52.9
v3 (Q4)	69.6	69.8	83.7	50.0	40.8
v5 (Q2)	79.2	85.2	85.7	75.1	56.3
v5 (Q4)	72.3	73.0	79.1	55.3	50.7
Human (SG)	93.5	76.7	83.7	59.1	54.8

On focal epileptiform, Mistral drops to ~40 while v3/v5 hold ~65–86. Even the human's certainty on the slowing classes is modest — those are genuinely hard to pin to an exact level.

Key points

- **Beats the external baseline everywhere.** v5 is above Mistral-7B in all five categories, decisively on the harder ones (e.g. Gen Epi 89 vs 75, Focal Non-epi 89 vs 76).
- **Near-human.** 87.6% vs the 89.8% human ceiling; on the epileptiform classes it already matches the human.
- **Captures confidence, not just the finding.** When the *exact* confidence level must match, Mistral collapses on focal epileptiform (F1 0.41) while v5 holds (0.67–0.77).
- **Generalizes.** The result holds on an unseen neurologist (Maria) and against a held-out second annotator — so it reflects real clinical reasoning, not fitting to one reader's phrasing.

- **Practical.** Runs on CPU; the smaller Q2 model is enough (same whole-report accuracy as Q4, at ~⅔ the size).

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v3 and v5 here are run with a consistency-guaranteeing decoding constraint (a grammar that forbids self-contradictory outputs). Full technical detail, all 10 prompt variants, the tested-and-rejected ideas, and per-dataset tables: all_prompts.md *· prompt texts:* `../prompts/`.